

VINAYAKA: Multilingual Audio-Visual Hate Speech Detection via Cross-Modal Fusion in Hyperbolic Space

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Abstract

In this work, we investigate the robustness of audio-visual (AV) cues for Multilingual Audio-Visual Hate Speech Detection (M-AVHSD). We propose **VINAYAKA**, a novel framework for M-AVHSD that relies solely on AV cues for decision-making. Along with WavLM and ImageBind, it employs Cross-modal Fusion in Hyperbolic Space (CFHS), which models the effective alignment between paralinguistic and behavioral AV cues to better capture hateful intent in temporal audio signals and corresponding visual scenes. Empirical results demonstrate that the proposed framework achieves state-of-the-art performance in both in-distribution and out-of-distribution settings across cross-lingual and cross-dataset evaluations. Furthermore, the results show that the framework remains robust to ASR error propagation.

Index Terms: Multimodal Hate Speech Detection, Audio-Visual Learning, Paralinguistic, Behavioral, Hyperbolic space

1. Introduction

With the widespread adoption of online media platforms, the production and consumption of short- and long-form video content have contributed to the propagation of hate speech. This makes hate speech detection (HSD) in video content being an important task for content moderation systems. In the real world, video content often involves code-mixing and code-switching across languages, especially in multilingual contexts such as India [1], where words are mixed between languages, or languages change across speakers. In such multilingual settings, the reliance on text-based representations for HSD is often unreliable. Moreover, hateful intent in online videos is not always conveyed through speech; visual cues may convey hostility, as gestures, symbols, or scenes, even when speech is neutral or absent. These challenges motivate us to investigate non-lexical audio-visual (AV) cues and to formulate multilingual audio-visual hate speech detection (M-AVHSD) that aims to detect hateful intent in videos without depending on language-specific textual cues.

Prior work treats hate speech detection as a purely text-classification task, focusing on lexical features [2] [3] [4]. Moving away from keyword-based methods, [5] explored sequential modeling, and [6] explored contextual representations by leveraging language-based transformer models. As summarized by Paula et al. [7], despite advances in modeling, automated content moderation systems remain heavily biased towards textual cues, with negligible consideration towards non-lexical cues. Several works extended hate speech detection to speech. [8] [9] [10] predominantly operate on Automatic Speech Recog-

nition(ASR) transcripts, eventually reducing speech to text. To address the limitations of text-only settings, researchers have begun incorporating paralinguistic features of speech. For instance, [11] demonstrated the importance of acoustic cues. Work on the ADIMA dataset shows that prosodic features like loudness and jitter are significant indicators of abusive behavior [11]. These findings show that the hostility is often found in the way someone speaks rather than just the words they use [12]. Beyond speech, efforts were made to explore static visual cues for hate or abusive content detection in the context of imagery, symbols, or memes [13] [14] demonstrating that hateful intent can also be conveyed through visuals in the absence of speech. More recently, multimodal approaches have been proposed that integrate audio, visual, and textual cues for HSD. Das et al. [15] provide a multimodal dataset and investigate the combination of audio, visual, and textual representations through cross-modal attention mechanisms. In the context of short-form videos, Maity et al. [16] introduce a multimodal benchmark for Hindi-English code-mixed content. Though these approaches [15] [16] [17] primarily rely on textual representation for decision-making, as textual cues are less susceptible to noise. Wang et al. [18] provided a cross-cultural perspective, benchmarking hate speech detection across English- and Chinese-language videos. Similarly, Ranjan et al. [19] explored multilingual HSD in synthetic audio. In an adjacent domain, Phukan et al. [20] made an effort to detect child-harmful content through audio-visual alignment. Further, for fine-grained HSD, Zhang et al. [21] provided the largest multimodal video dataset to date.

Despite recent strides in multimodal hate speech detection (MHSD) [17] [22] [23], several questions remain unanswered. The decision-making process in multimodal settings is dominated by textual representations. As a consequence, such systems remain highly susceptible to ASR error propagation [24], leading to degraded performance under cross-lingual, code-mixed, and domain shifts. Works increasingly included audio and visual signals but treated them as auxiliary. Their behavioral and paralinguistic characteristics, such as prosody, timing, and facial dynamics, which are main indicators of hateful intent, have not been investigated. Furthermore, the extent to which current frameworks generalize to multilingual and out-of-distribution settings, where textual cues are unreliable, is not investigated. Most prior approaches operate within Euclidean representation spaces. This leaves it unclear whether the inherently hierarchical structure of behavioral cues can be effectively modeled in an alternative geometric representations, like hyperbolic space.

To address this gap, we investigate the extent to which AV cues can support robust M-AVHSD without relying on textual representations across multilingual and out-of-distribution con-

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ditions. *We hypothesize that AV cues, due to their paralinguistic and behavioral nature, generalize better across languages and datasets than text-centric approaches. We posit that the inherently hierarchical structure of these cues can be better modeled through cross-modal fusion in hyperbolic space, enabling better alignment of temporal audio and corresponding video associated with hateful intent.* Our experiments, comparing state-of-the-art (SOTA) [17] [16] [18], confirm the validity of our hypothesis. To our end, we propose a novel framework, **VINAYAKA**, that performs cross-modal fusion in hyperbolic space (CFHS) between the audio and visual modalities for better AV alignment and interaction between temporal audio signals and their corresponding visual scenes. **VINAYAKA** achieves superior performance compared to audio-only, video-only, and other Euclidean fusion baselines; therefore, we report the state-of-the-art (SOTA) work for M-AVHSD.

To summarize, our contributions are fourfold:

- To our end, we are the first to investigate the robustness of paralinguistic and behavioral AV cues for M-AVHSD via CFHS.
- We propose **VINAYAKA**, a novel framework for M-AVHSD. Through CFHS, it effectively models and aligns the interaction between temporal audio and corresponding visuals to capture hateful intent.
- We demonstrate that AV features generalize more effectively than text-centric representations across code-mixed, cross-lingual, and out-of-distribution conditions.
- Our results show that **VINAYAKA** achieves the **state-of-the-art (SOTA)** performance in in-distribution and in out-of-distribution settings, establishing a baseline for M-AVHSD.

We will release the code and the trained models after the double-blind review process.

2. Methodology

In this section, we first discuss the pre-trained models (PTMs) used for feature extraction, then the methodologies for unimodal and multimodal baselines with various fusion methods, and our proposed framework **VINAYAKA**.

2.1. Feature Extraction

We select WavLM [25] and the ImageBind visual encoder [26] to obtain speech embeddings from raw waveforms and visual embeddings from sampled video frames, respectively. WavLM is pre-trained on 94K hours of unlabeled speech data using techniques such as masked prediction and utterance mixing. The pre-training corpus spans diverse acoustic conditions, encompassing YouTube videos, audiobooks, and political speeches, enabling WavLM to learn robust paralinguistic features, such as prosodic and speaker-dependent features. ImageBind is a multimodal model pre-trained to map six modalities, including vision and audio, into a unified embedding space. In our setup, we pass uniformly sampled video frames into the frozen visual encoder, obtaining tubelet-based spatiotemporal embeddings that capture motion patterns, scene-level, and object-level understanding present in the video. Both architectures are kept frozen to obtain the embeddings for downstream processing.

2.2. Modeling

2.2.1. Unimodal

In unimodal settings, both modalities follow a similar downstream network: feature extraction from PTMs, a CNN-based architecture, and a Fully Connected Network (FCN). The extracted features are then passed through the two 1D-CNN layers (64 and 128 filters, kernel size 3), each followed by max pooling (pool size = 2). The resulting features are then flattened and passed through the FCN block with a dense layer of 128 neurons. The output layer with a softmax activation function produces the class probabilities.

2.2.2. Multimodal Baseline Audio-Visual Fusion Methods:

We also implement multiple fusion strategies, including feature concatenation (CON), Euclidean cross-attention (ECA), Möbius addition (MA), and cross-modal fusion in hyperbolic space (CFHS) as in our **VINAYAKA** model. For comparison with state-of-the-art models, we use the official implementations of MM-HSD [17] and ToxVidLM [16]. As no source code is available for the MultiHateClip [18], we replicate its architecture strictly following the methodologies described in the original paper. All the models are trained under the same experimental settings. In CON, embeddings from WavLM and ImageBind are passed through separate CNNs and pooled pipelines, as in unimodal settings, then concatenated and fed to the FCN block for classification. For ECA and MA, the same pipeline is used until fusion and after fusion. For ECA, a cross-modal attention mechanism is applied, and the attended features are fused. For MA, the features are projected to separate hyperbolic Poincaré balls, then fused via the Möbius addition operation.

2.3. VINAYAKA

We propose **VINAYAKA**, a multimodal framework for M-AVHSD that effectively aligns and fuses behavioral cues across audio and visual modalities. **VINAYAKA** models the hierarchical dependencies between the audio and video representations through the CFHS mechanism within the negative curvature restrictions of the hyperbolic manifold. CFHS addresses the gap of ECA by enabling modeling of hierarchical audio-visual (AV) interactions, which are difficult to capture using Euclidean representations alone. An overview of the **VINAYAKA** architecture is shown in Figure 1

Starting with feature extraction, the extracted features from WavLM $\mathbf{E}^{(a)} \in \mathbb{R}^{T_a \times d_a}$, and ImageBind, $\mathbf{E}^{(v)} \in \mathbb{R}^{T_v \times d_v}$ are passed through the CNN block with the same architectural details defined in the unimodal stage, which are further flattened, resulting features are denoted as $\mathbf{H}^{(a)} \in \mathbb{R}^{T'_a \times d}$ and $\mathbf{H}^{(v)} \in \mathbb{R}^{T'_v \times d}$.

Further, these Euclidean features are then projected to the hyperbolic Poincaré ball $\mathbb{D}_c^d$ where the curvature is $-c$: $\mathbb{D}_c^d = \{\mathbf{x} \in \mathbb{R}^d \mid c\|\mathbf{x}\|^2 < 1\}$. After this, we apply the exponential map at the origin:

$$\mathbf{Z}^{(a)} = \exp_0^c(\mathbf{H}^{(a)}), \quad \mathbf{Z}^{(v)} = \exp_0^c(\mathbf{H}^{(v)}) \quad (1)$$

where $\exp_0^c(\mathbf{x}) = \tanh(\sqrt{c}\|\mathbf{x}\|) \frac{\mathbf{x}}{\sqrt{c}\|\mathbf{x}\|}$.

Then, for CFHS, we define Queries $\mathbf{Q}^{(m)}$, Keys $\mathbf{K}^{(m)}$, and Values $\mathbf{V}^{(m)}$ in $\mathbb{D}_c^d$ for each modality $m \in \{a, v\}$, through Möbius linear transformations of $\mathbf{Z}^{(m)}$. CFHS utilizes the hyperbolic distance $d_c(\cdot, \cdot)$ to compute cross-modal attention weights. **Attention Weights** $\alpha_{ij}^{a \rightarrow v}$ denoting audio-to-visual $\alpha_{ij}^{v \rightarrow a}$ denot-

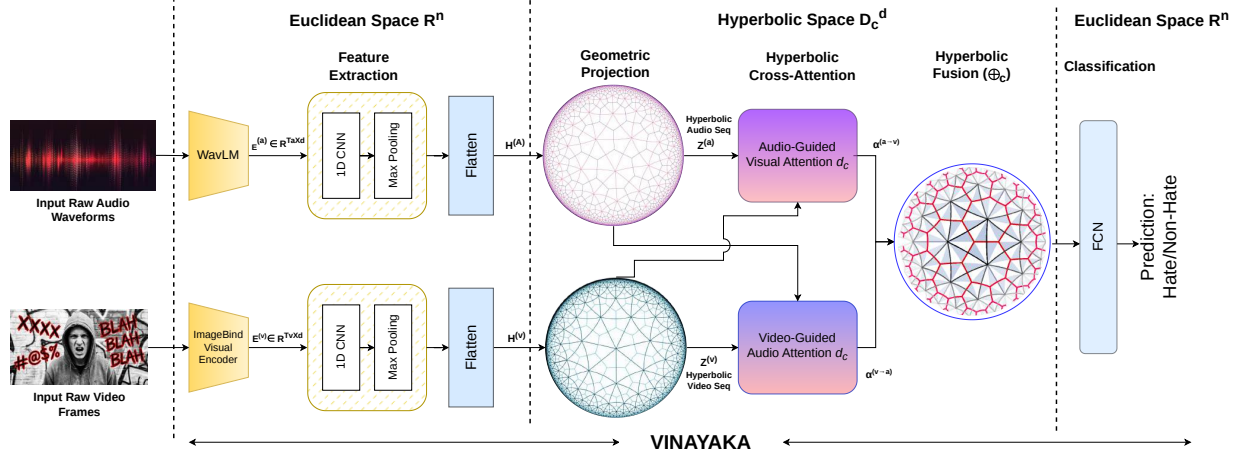


Figure 1: Overview of the proposed VINAYAKA architecture.

ing visual-to-audio are computed as:

$$\alpha_{ij}^{m \rightarrow n} = \frac{\exp(-d_c(\mathbf{Q}_i^{(m)}, \mathbf{K}_j^{(n)}))}{\sum_{j'} \exp(-d_c(\mathbf{Q}_i^{(m)}, \mathbf{K}_{j'}^{(n)}))} \quad (2)$$

where $(m, n) \in \{(a, v), (v, a)\}$

Möbius Operations. For feature aggregation of attended weights computed in Eq. (2) with corresponding Values $\mathbf{V}$ in a hyperbolic manifold, we employ Möbius scalar multiplication ($\otimes_c$) and Möbius addition ($\oplus_c$). The attended representations are computed as:

$$\mathbf{O}_i^{a \rightarrow v} = \bigoplus_j (\alpha_{ij}^{a \rightarrow v} \otimes_c \mathbf{V}_j^{(v)}), \mathbf{O}_i^{v \rightarrow a} = \bigoplus_j (\alpha_{ij}^{v \rightarrow a} \otimes_c \mathbf{V}_j^{(a)}) \quad (3)$$

The bidirectional representations from Eq. (3) are fused using Möbius addition: $\mathbf{O}^{(H)} = \mathbf{O}^{a \rightarrow v} \oplus_c \mathbf{O}^{v \rightarrow a}$. For further classification, the fused representations are projected back to the Euclidean space. For projection, we employ the logarithmic map:

$$\mathbf{O}^{(E)} = \log_0^c(\mathbf{O}^{(H)}) = \frac{1}{\sqrt{c}} \tanh^{-1}(\sqrt{c} \|\mathbf{O}^{(H)}\|) \frac{\mathbf{O}^{(H)}}{\|\mathbf{O}^{(H)}\|} \quad (4)$$

The Möbius addition and Möbius scalar multiplication operations used to perform feature aggregation and cross-modal fusion in Eq. (3), (4) are defined for $\mathbf{x}, \mathbf{y} \in \mathbb{D}_c^d$ and $r \in \mathbb{R}$ as follows:

$$\mathbf{x} \oplus_c \mathbf{y} = \frac{(1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c\|\mathbf{y}\|^2)\mathbf{x} + (1 - c\|\mathbf{x}\|^2)\mathbf{y}}{1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c^2\|\mathbf{x}\|^2\|\mathbf{y}\|^2} \quad (5)$$

$$r \otimes_c \mathbf{x} = \tanh(r \tanh^{-1}(\sqrt{c}\|\mathbf{x}\|)) \frac{\mathbf{x}}{\sqrt{c}\|\mathbf{x}\|} \quad (6)$$

Hyperbolic Distance. We utilize the geodesic distance on the manifold $\mathbb{D}_c^d$ to measure the similarity between Query and Key representations for the attention weights,

$$d_c(\mathbf{x}, \mathbf{y}) = \frac{2}{\sqrt{c}} \tanh^{-1}(\sqrt{c}\|-\mathbf{x} \oplus_c \mathbf{y}\|) \quad (7)$$

where $\oplus_c$ denotes the Möbius addition.

The resulting vector is passed through a fully connected network with a softmax layer for obtaining class probabilities.

3. Experiments

3.1. Dataset

HateMM [15]: It contains 1083 videos (431 Hateful and 652 Non-Hateful) of approximately 43 hours and 144K frames curated from the BitChute social video-hosting platform. It serves as a stable English-only source domain with clear AV hate cues.

ToxCMM [16]: It is a code-mix of Hindi-English language from a conversation dataset with 4021 utterances curated from 931 videos scraped from YouTube. We have followed the official binary split, labeling 'Toxic' as 'Hateful' and 'Non-Toxic' as 'Non-Hateful'. This dataset supports our study, as ASR and textual representations are unreliable in code-mixed settings.

MultiHateClip [18]: It comprises 2000 samples curated from platforms such as YouTube and Bilibili. It contains English (MHCE) and Chinese (MHCC) videos labelled as 'Hateful', 'Offensive', and 'Normal'. We have followed 'Hateful' and 'Offensive' samples as 'Hateful', and 'Normal' labeled samples as 'Non-Hateful'. It serves our purpose as a cross-cultural and cross-domain dataset.

3.2. Training Details

For HMM (HateMM), ToxCMM, MHCE, and MHCC, we adopted 5-fold cross-validation splits across all experiments. For cross-dataset evaluation, all models were trained on the source dataset and evaluated directly on the target dataset, without target-domain finetuning. To obtain ASR transcripts, we used Whisper [27] and PaddleOCR [28] for OCR wherever required for text-dependent baselines. For all models, including unimodal and multimodal baselines, we use the Adam optimizer with an initial learning rate of $1e^{-4}$, a batch size of 16, a Binary cross-entropy loss function, and early stopping based on the validation macro-F1 score. The reported accuracy and macro-F1 scores are averaged across folds. To show the scalability of VINAYAKA, we follow zero-intervention policy for all baselines and their respective modalities, unlike prior works where [16] that relied on manual transcript cleaning.

3.3. Experimental Results

We present a robust evaluation of VINAYAKA via 1) In-domain ablation—establishing the necessity of CFHS, 2) Cross-dataset

Table 1: *Performance across In-domain and Cross-dataset Evaluation. Acc and F1 refer to Accuracy and Macro-Average F1, respectively. The abbreviations given are: Audio only (A), Video only (V), Concatenation (CON), Euclidean Cross-Attention (ECA), Möbius Addition (MA), OURS (VINAYAKA).*

Model	HateMM		MH-En		MH-Ch		ToxCMM	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
In-Distribution Evaluation								
A	.798	.786	.648	.636	.642	.629	.781	.774
V	.815	.804	.664	.652	.661	.649	.768	.761
CON	.836	.826	.708	.695	.701	.689	.807	.797
ECA	.861	.848	.748	.735	.755	.742	.848	.835
MA	.883	.871	.782	.779	.739	.725	.835	.825
OURS	.914	.901	.849	.841	.847	.831	.892	.885
Out-of-Distribution / Cross-Dataset Evaluation								
<i>Training Dataset: HateMM (HMM)</i>								
MM-HSD [17]	.878	.874	.623	.602	.416	.422	.597	.581
MHC [18]	.809	.795	.697	.681	.511	.506	.556	.542
TVLM [16]	.833	.821	.769	.761	.497	.480	.632	.641
OURS	.914	.901	.808	.789	.563	.551	.724	.712
<i>Training Dataset: MultiHateClip-En (MHCE)</i>								
MM-HSD [17]	.733	.717	.815	.801	.507	.483	.622	.609
MHC [18]	.711	.727	.810	.790	.595	.541	.639	.622
TVLM [16]	.755	.729	.833	.812	.520	.503	.641	.622
OURS	.797	.786	.849	.841	.626	.619	.715	.706
<i>Training Dataset: MultiHateClip-Ch (MHCC)</i>								
MM-HSD [17]	.538	.540	.518	.492	.834	.820	.555	.527
MHC [18]	.446	.425	.579	.562	.800	.780	.411	.392
TVLM [16]	.528	.515	.299	.330	.609	.576	.527	.508
OURS	.566	.557	.619	.616	.847	.831	.621	.612
<i>Training Dataset: ToxCMM</i>								
MM-HSD [17]	.602	.581	.672	.657	.410	.390	.828	.819
MHC [18]	.596	.577	.649	.633	.555	.534	.789	.771
TVLM [16]	.558	.522	.700	.683	.555	.521	.823	.842
OURS	.726	.712	.709	.693	.585	.577	.892	.885

Table 2: *Geometric Ablation of CFHS under Different Curvature Settings.*

Curvature (c)	HMM	MHCE	MHCC	TCMM
$c = 0$ (Euclidean)	0.877	0.809	0.803	0.868
$c = +1$ (Spherical)	0.801	0.768	0.742	0.759
$c = -1$ (Hyperbolic)	0.901	0.841	0.831	0.885

generalization across languages and domains to show robustness in out-of-distribution 3) A Geometric ablation in Table 2 proving the impact of manifold curvature. Table 1 presents the results of In-domain and cross-dataset evaluation.

Unimodal Baselines: Across HMM, MHCE, and MHCC, the visual modality surpasses the audio modality by a small margin, as these datasets contain explicit visual cues such as gestures, memes, symbols, and scenes that directly communicate hateful intent. In contrast, on TCMM, the audio modality slightly outperforms the visual modality due to the presence of explicit vocal aggression. We observed that neither of these modalities alone is sufficient for consistent hate detection across linguistic and visual domain shift.

Multimodal Baselines: To benchmark the effectiveness of VINAYAKA, we also compared it against various fusion strategies such as CON, ECA, and MA as seen in Table 1. VINAYAKA outperforms all the multimodal baselines in the In-distribution settings, demonstrating the effectiveness of the CFHS between the audio-visual modalities. Comparing ECA

and MA shows complementary behavior. MA surpasses ECA, suggesting that curvature-aware aggregation offers representational advantages. ECA slightly outperforming MA suggests negative curvature is insufficient without precise alignment. The performance gains in VINAYAKA suggest that integrating CFHS helped to capture structured AV interactions more effectively, than either modality alone. To validate the necessity of hyperbolic geometry, we conducted a geometric ablation study comparing Spherical ($c > 0$), Euclidean ($c = 0$), and Hyperbolic ($c < 0$) manifolds.

Cross-Dataset Evaluation: While training on HMM, VINAYAKA achieves 0.901 F1 in-distribution. The transfer to MHCE shows a moderate drop to F1 0.789, indicating visual overlap in the political themes across the English datasets. The transfer to TCMM results in a drop to 0.712 F1, suggesting differences in symbolic, monologued content and the acoustics of code-mixed utterances. When trained on MHCE, VINAYAKA achieves in-distribution F1 of 0.841. The transfer to HMM and TCMM shows F1 of 0.786 F1 and 0.706 F1, respectively, indicating that conversational hate patterns transfer effectively across English datasets and moderately low in cross-lingual. The transfer to MHCC resulted in an F1 score of 0.609, demonstrating the challenges across cross-cultural and cross-lingual transfer. In MHCC as the training set, our model scores an in-distribution of 0.831 F1. On the same, F1 scores of 0.616, 0.612, and 0.557 across MHCE, HMM, and TCMM, respectively, demonstrate the hardest transfers. Training on TCMM emphasizes conversational behavioral and prosodic cues. VINAYAKA, leveraging acoustic and visual behavioral cues, yields a F1 of 0.885. The drops to F1 of (HMM), 0.693 (MHCE), and 0.577 (MHCC) indicate moderate robustness under distribution shift.

Under cross-lingual transfer, text-dependent baselines show significant drops, highlighting ASR error propagation due to their reliance on lexical cues. While VINAYAKA relies on paralinguistic and behavioral AV cues, it demonstrates a comparatively stable performance. Also, the WavLM’s prosodic features and ImageBind’s high-level semantic representations exhibit a lower sensitivity to language variation. This is consistent with our motivation: *text-based systems are highly susceptible to ASR error propagation and degrade under cross-lingual and domain shift*. Furthermore, across all cross-dataset experiments, VINAYAKA’s CFHS, via its structured cross-modal alignment, consistently shows smaller drops and performs better than other fusion methods. This indicates improved stability in out-of-distribution settings. These findings suggest that paralinguistic and behavioral AV cues, when effectively modeled via CFHS, provide superior stability than text-centric baselines in cross-lingual, cross-dataset settings.

4. Conclusion

In this study, we investigated the robustness of AV cues for M-AVHSD. We demonstrated that AV cues solely, without relying on textual cues shows stable and improved performance in in-distribution and out-of-distribution settings, compared to other text-centric baselines. This highlights the critical advantage of avoiding ASR error propagation. Our novel framework, VINAYAKA, that aligns paralinguistic and behavioral AV cues from WavLM and ImageBind via cross-modal fusion in hyperbolic space, effectively model cross-modal interactions and achieve state-of-the-art performance on M-AVHSD. Empirically, our framework demonstrates improved stability in cross-lingual and cross-dataset transfer scenarios. Overall, our find-

ings support our hypothesis that paralinguistic and behavioral AV cues are more robust than text-centric representations in multilingual and out-of-distribution settings.

5. Generative AI Use Disclosure

AI-assisted tools were utilized solely to improve the grammar, clarity, and presentation of the manuscript. They did not contribute to the development of scientific concepts, data analysis, result generation, or interpretation of the findings. The authors assume full responsibility for the accuracy, validity, and integrity of the research.

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